



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science
Faculty of Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2025-2026

Section: [EE]

Project Report On

Project Name: **Lawyer Management System**

Supervised By

DR. MD. IFTEKHARUL MOBIN

Submitted By:

Name	Id
AHNAF AZMAIN ANIK	24-59604-3
AHSAN KABIR ARANNA	24-60365-3
MD.NAHIAN HOSSAIN SIAM	24-59572-3
NASHID JAMIL KANCHON	23-54572-3

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total =
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

Title

Lawyer Management System: A Role-Based Digital Platform for Efficient Legal Case Handling and Administration

Introduction

The rapid digital transformation of service-based industries has increased the need for efficient, secure, and automated management systems. Legal case handling, traditionally dependent on manual paperwork and human coordination, often suffers from delays, miscommunication, and lack of transparency.

To address these issues, this project presents a **C# Windows Forms-based Lawyer Management System integrated with MS SQL Server**. The system is designed to streamline interactions between clients, lawyers, and administrative roles through a structured digital platform.

The application supports **role-based access control**, including Client, Lawyer, Admin, and SuperAdmin. Each role has dedicated functionalities that ensure proper workflow management, security, and accountability.

The main objective of this system is to:

- Digitize legal case management
- Improve communication between stakeholders
- Ensure transparency in case tracking and reviews
- Automate administrative tasks such as case assignment, status updates, and payment/commission handling

By integrating database-driven operations with an intuitive GUI, the system enhances efficiency and reduces manual workload in legal service management.

Case Study

In traditional legal service environments, clients often face difficulties in finding suitable lawyers, tracking case progress, and receiving timely updates. Lawyers, on the other hand, manage multiple cases manually, leading to inefficiency and missed deadlines. Administrative staff also struggle with organizing records and ensuring fair distribution of cases.

This system addresses these challenges through a structured case-based workflow:

Case Flow Scenario:

1. Client Registration & Login

- Clients create accounts and access available lawyers.
- They can browse lawyer profiles, ratings, and reviews.

2. Hiring a Lawyer

- Clients submit a case request and select a lawyer.
- The request is stored in the database under a pending status.

3. Lawyer Response

- Lawyers receive case requests in their dashboard.
- They can accept or reject cases based on availability.

4. Case Processing

- Once accepted, the case status is updated to “In Progress”.
- Lawyers update case progress until completion.

5. Admin Monitoring

- Admin oversees all cases, ensures proper assignment, and resolves disputes.
- Admin can also manage lawyer performance and client complaints.

6. SuperAdmin Control

- SuperAdmin has full system authority.
- Can manage admins, remove underperforming lawyers, and control financial operations like commissions.

Common Functionalities

Authentication System

- Secure login for all users (Client, Lawyer, Admin, SuperAdmin)
- Role-based redirection to respective dashboards
- Password-protected access using database verification

3.2 User Management

- Registration and profile management for clients and lawyers
- Admin-controlled verification of lawyers
- SuperAdmin authority over all user accounts

3.3 Case Management

- Clients can create and submit legal cases
- Lawyers can accept, process, and complete cases
- Real-time case status updates stored in MS SQL Server

3.4 Dashboard System

- Separate dashboards for each role:
 - Client Dashboard: hire lawyers, view case status, give feedback
 - Lawyer Dashboard: manage assigned cases
 - Admin Dashboard: monitor system activity
 - SuperAdmin Dashboard: full system control

3.5 Review & Rating System

- Clients can rate lawyers after case completion
- Ratings help future clients choose appropriate lawyers
- Improves system transparency and accountability

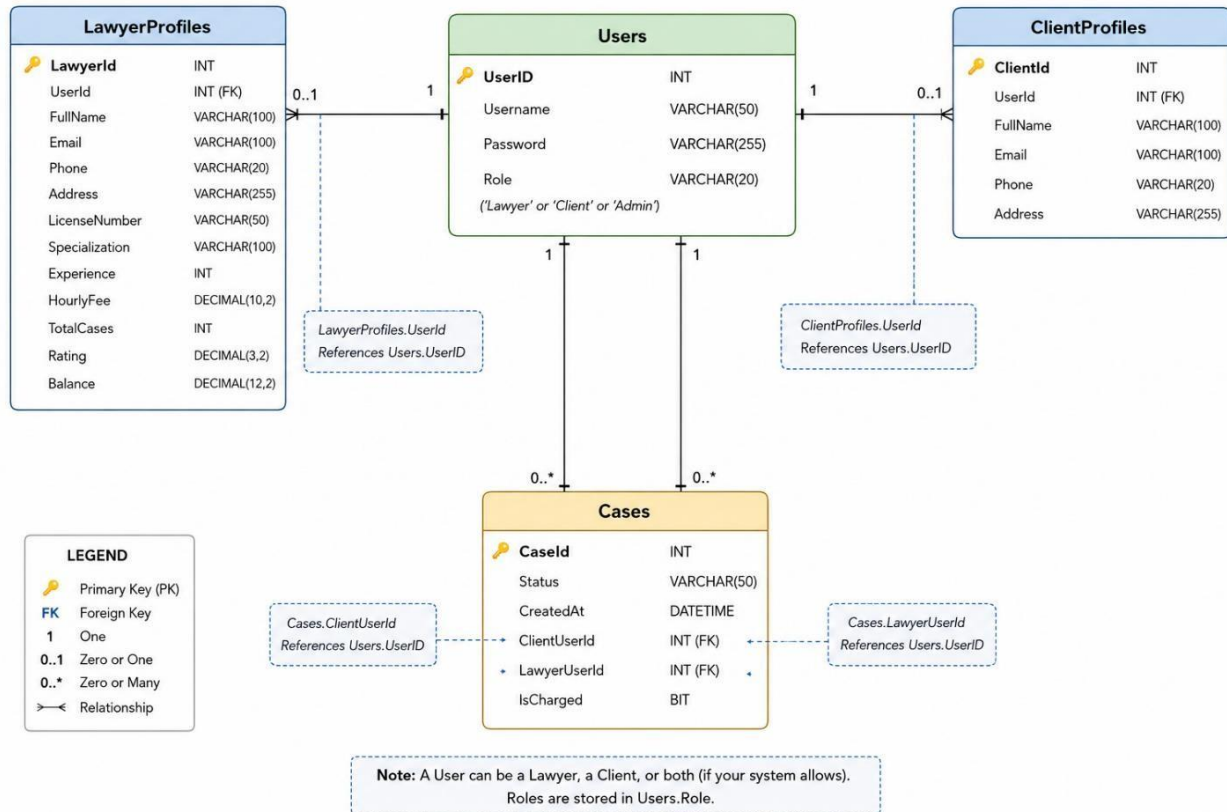
3.6 Notification & Status Tracking

- Case status tracking (Pending, Accepted, In Progress, Completed)
- Basic notification system for updates between users

3.7 Administrative Control

- Admin can manage cases and users
- SuperAdmin can manage admins, delete records, and control system-level settings

Database Schema



Database Query

Users table

```
CREATE TABLE Users (
  UserID INT IDENTITY(1,1) PRIMARY KEY, Username
  VARCHAR(50) NOT NULL UNIQUE, Password
  VARCHAR(255) NOT NULL,
  Role VARCHAR(20) NOT NULL CHECK (Role IN ('Lawyer', 'Client', 'Admin'))
);
```

Lawyer Profiles

```
CREATE TABLE LawyerProfiles (
  LawyerId INT IDENTITY(1,1) PRIMARY KEY,
  UserId INT UNIQUE, -- one-to-one with Users
```

```
  FullName VARCHAR(100),
  Email VARCHAR(100),
```

Phone VARCHAR(20), Address
VARCHAR(255),

LicenseNumber VARCHAR(50), Specialization
VARCHAR(100), Experience INT,
HourlyFee DECIMAL(10,2),
TotalCases INT DEFAULT 0,
Rating DECIMAL(3,2),
Balance DECIMAL(12,2) DEFAULT 0,

FOREIGN KEY (UserId) REFERENCES Users(UserID) ON
DELETE CASCADE
);

Client Profiles

CREATE TABLE ClientProfiles (
ClientId INT IDENTITY(1,1) PRIMARY KEY,
UserId INT UNIQUE, -- one-to-one

FullName VARCHAR(100),
Email VARCHAR(100), Phone
VARCHAR(20), Address
VARCHAR(255),

FOREIGN KEY (UserId) REFERENCES Users(UserID) ON
DELETE CASCADE
);

Cases

CREATE TABLE Cases (
CaseId INT IDENTITY(1,1) PRIMARY KEY,

Status VARCHAR(50) DEFAULT 'Open', CreatedAt
DATETIME DEFAULT GETDATE(),

ClientUserId INT,

LawyerUserId INT,

IsCharged BIT DEFAULT 0,

FOREIGN KEY (ClientId) REFERENCES Users(UserID), FOREIGN
KEY (LawyerUserId) REFERENCES Users(UserID)

);

Transition Table

1. Login process



The screenshot shows a web application window titled "Login - Lawyer Management System". The main heading is "Lawyer Management System" with a logo of a golden scale of justice to its right. Below the heading is the word "Login". There are two input fields: "Username:" with the value "ahsan" and "Password:" with masked characters "••••". Below the password field is a red "Login" button. At the bottom, there is a link "Don't have an account" and a cyan "Register" button.

2. Register process

Register_Lawyer Management system

Register

Username

Password

Confirm Password

Role

3. Client Profile Setup

Profile setup

Setup Client Account

Full Name

Email

Phone

Address

4. Lawyer Profile Setup

Profile Setup

Full name

akash khan

License Number

1390

Email

akash@gmail.com

Experience

4

Phone

23454565554

Hourly Fee

500

Address

north badda

Specialization

Cyber Law

Save Profile

5. Client Dashboard

Client Dashboard

Welcome Client

log out

Search Lawyer

Specialization

Cyber Law

Max Hourly Fee

Below 1000

Minimum Experience

3+ Years

Search

Lawyers

	UserId	FullName	Specialization	Experience	HourlyFee	
▶	24	akash khan	Cyber Law	4	500.00	Hire
*						

My Case

	CasId	LawyerName	Status	CreatedAt
*				

Rate the Lawyer

	CasId	LawyerUser	LawyerName	CreatedAt	
▶	9	9	soulgoof ...	5/1/2026 ...	▼

Submit

8. Super Admin Dashboard

Superadmin Dashboard

×

Welcome to Super Admin Dashbord

Edit Admins

	Userld	Username
▶	18	admin
✱		

Delete

Add Admin

Username

Password

Add

Charge Lawyers

	Userld	Username	CompletedCases	UnchargedCases	Balance
▶	9	soul	4	0	0.00
	23	kyle	0	0	0.00
	24	akash	0	0	0.00

Charge

Update Profile

Username

super

Password

1234

Update